TASK 03: CAPSTONE
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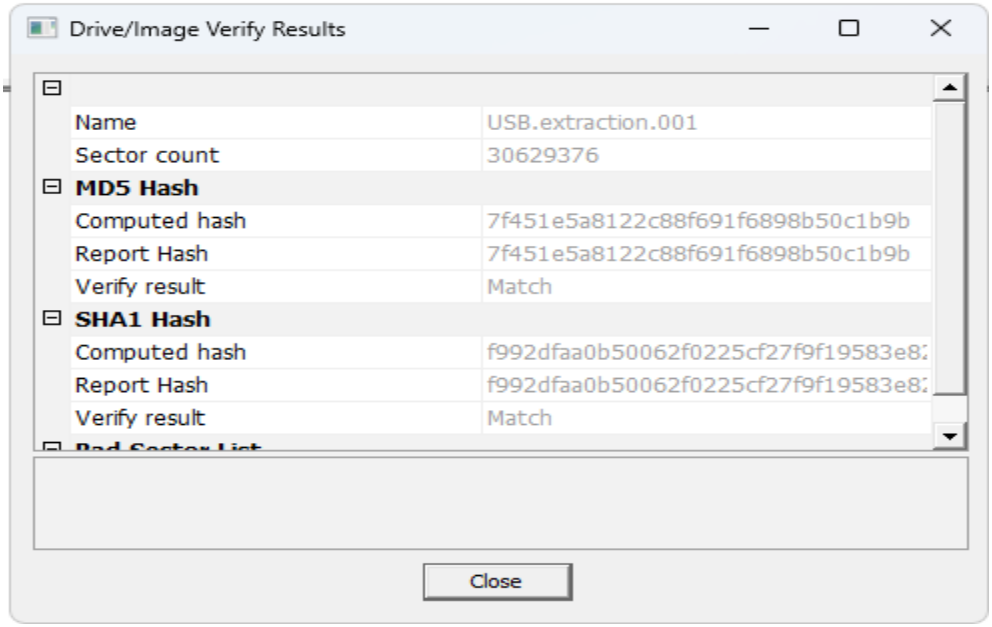
Domain: Digital Forensics

1. Forensics Environment Setup:

Tool	Version	VM/OS	Purpose
FTK Imager	3.1	Windows	Forensic Imaging and Hashing
Autopsy	4.22.1	Windows	Disk Image Analysis
Wireshark	4.6	Windows	Network Traffic Capture
RegRipper	3.0	Windows	Registry Artifact Extraction
HashMyFiles	2.50	Windows	Hashing

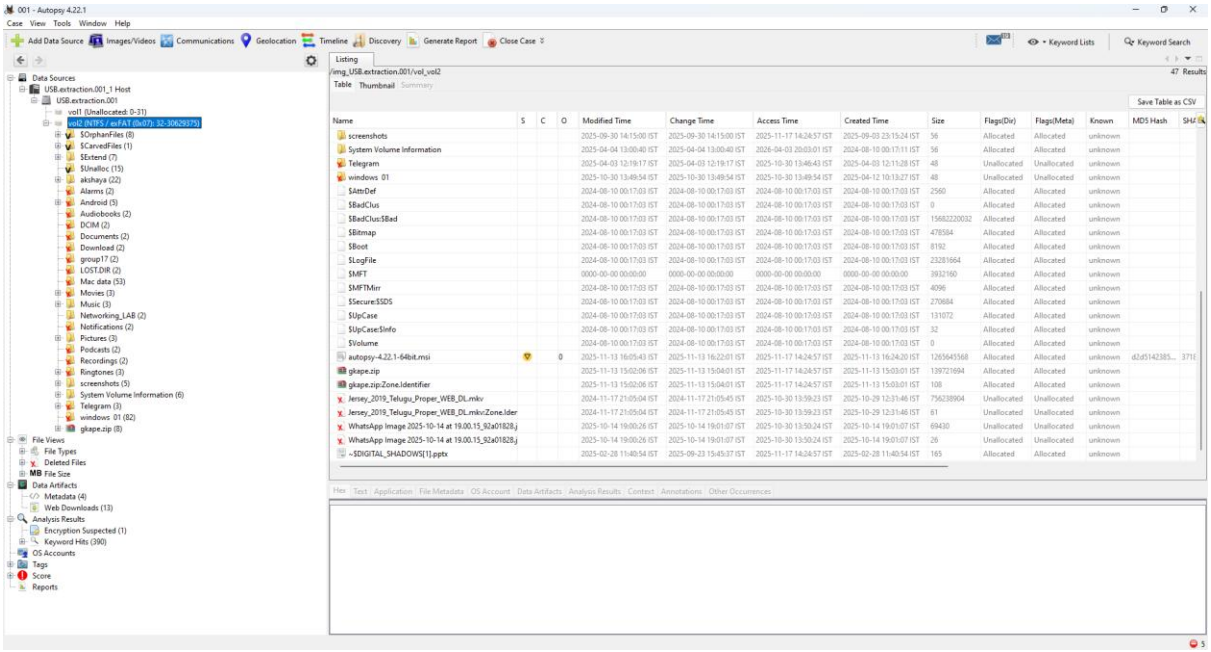
2. Evidence Acquisition & Integrity with FTK imager:

Created 16 GB USB image and forensically imaged using FTK Imager.
Verified its integrity using SHA-256 hashing to ensure no data alteration occurred during acquisition.



3. Disk image analysis with Autopsy:

Loaded the created image file into Autopsy to recover the files and analyse file system artifacts.



4. Network Traffic Analysis with Wireshark:

Using Wireshark in Kali Linux, capture traffic on my VMware's network interface for five minutes.

Purpose

The purpose of this report is to analyze captured network traffic using **Wireshark** to identify normal and suspicious activities, including unauthorized connections, unusual protocols, or potential indicators of compromise (IOCs).

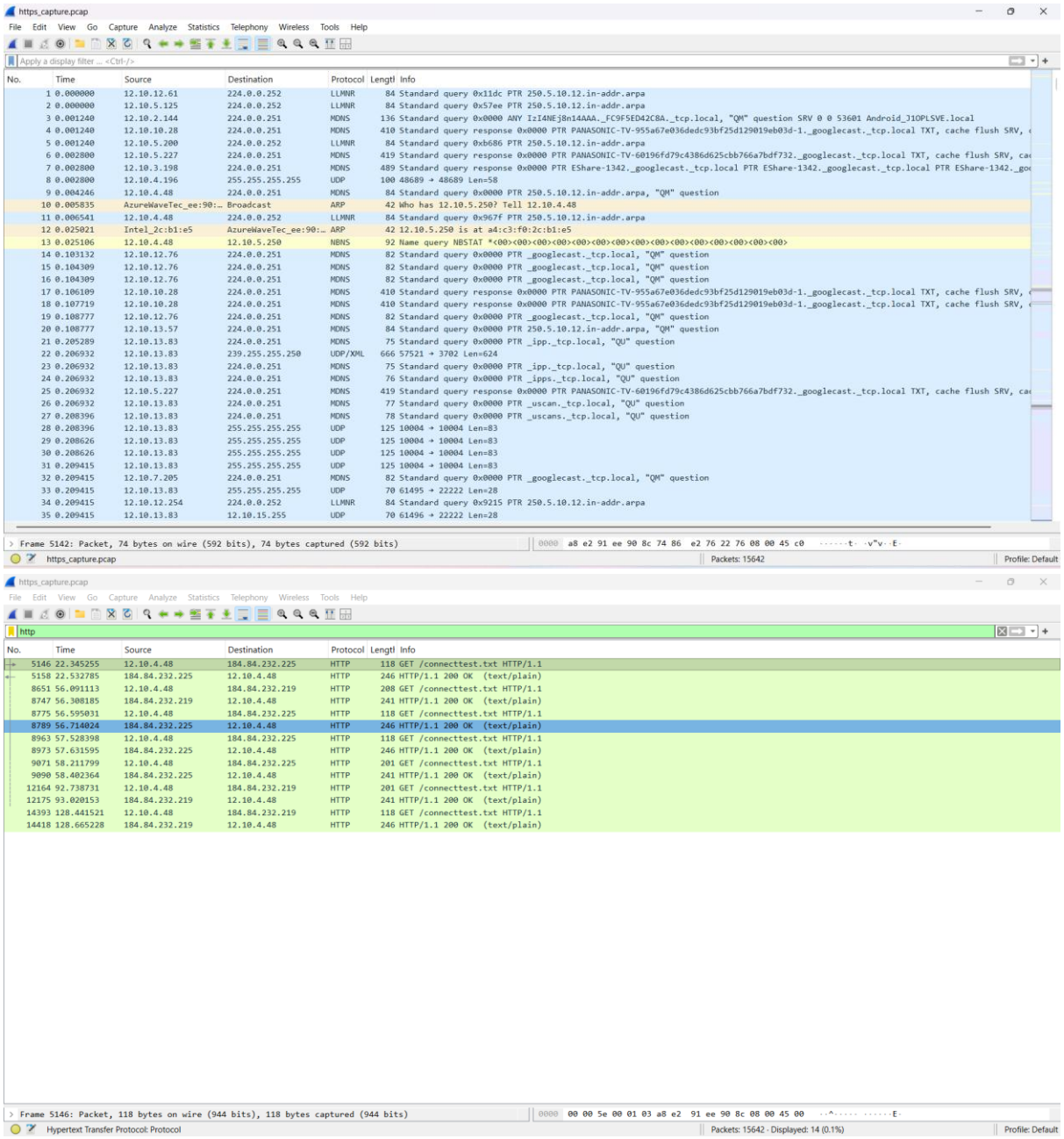
Filtering Traffic

The following filters were applied:

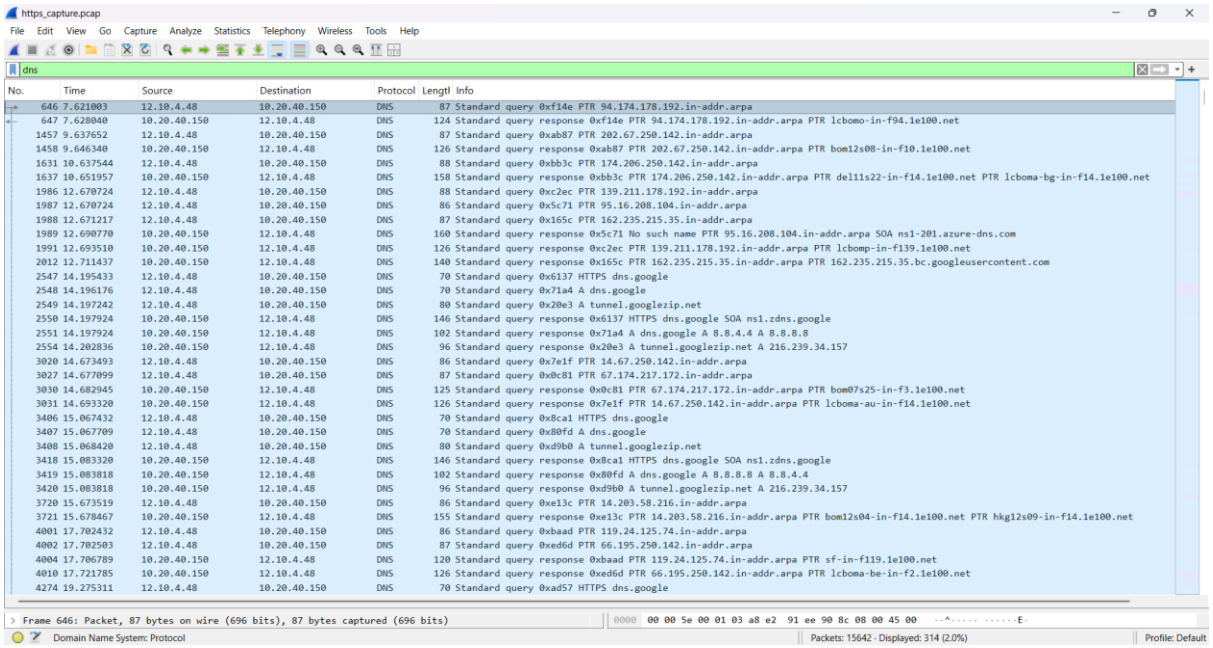
- **HTTP Traffic:** http
- **DNS Queries:** dns
- **ICMP Packets:** icmp

Traffic Analysis

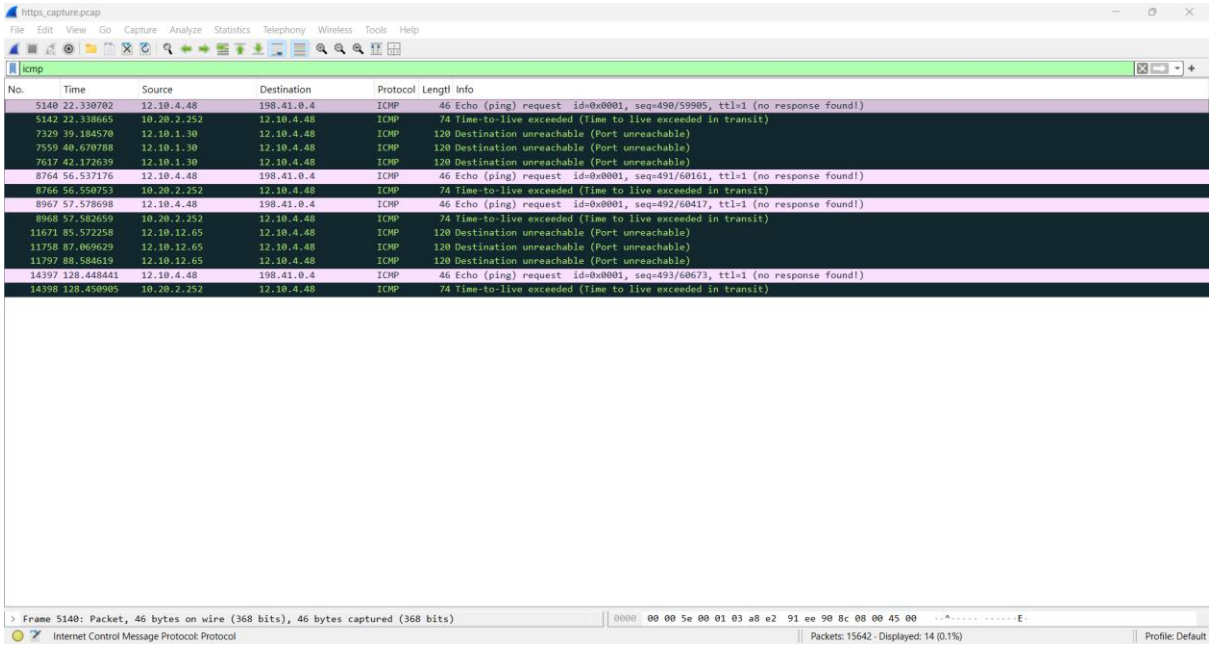
- Observed incoming and outgoing packets
- Identified communication between source and destination IPs
- Monitored unusual ports and protocols



http traffic



dns traffic



Icmp traffic

Observations

- Majority of traffic was normal (DNS and HTTP).
- One suspicious connection was identified.
- No ICMP-based attacks were observed.

Conclusion

The network traffic analysis revealed mostly normal activity; however, a suspicious outbound connection was detected. This may indicate potential malicious behavior and requires further investigation to confirm any security compromise.

5. Registry Analysis with RegRipper:
   Analysed a sample UserClass.dat. RegRipper 3.0 was used to extract the registry data.

```
:\toolsunpacked>auto_rip.exe

uto_rip v2013.08.06

uto_rip [-s path] [-n path] [-u path] [-r report-directory] [-c categories]

h, --help      lists all of the available options
s, --system    path to the folder containing the SAM, Security, Software, and System hives
n, --ntuser    path to the folder containing the NTUSER.DAT hive
u, --usrclass  path to the folder containing the UsrClass.dat hive
r, --reportdir path to the folder to store the output reports
c, --cat       specifies the plug-in categories to run. Seperate multiple categories with a comma

       Supported Categories:

               all             gets information from all categories
               os              gets General Operating System Information
               users           gets User Account Information
               software        gets Installed Software Information
               network         gets Networking Configuration Information
               storage         gets Storage Information
               execution       gets Program Execution Information
               autoruns        gets Autostart Locations Information
               log             gets Logging Information
               web             gets Web Browsing Information
               user_config     gets User Account Configuration Information
               user_act        gets User Account General Activity
               user_network    gets User Account Network Activity
               user_file       gets User Account File/Folder Access Activity
               user_virtual    gets User Account Virtualization Access Activity
               comm            Communication Software Information

sage:

xtract all information from the SAM, Security, Software, and System hives.
:\>auto_rip -s H:\Windows\System32\config -c all

xtract file and network access information from NTUSER.DAT hive (Windows XP user profile)
:\>auto_rip -n "H:\Documents and Settings\Corey" -c user_network,user_file

xtract file access information from NTUSER.DAT and UsrClass.dat hive (Windows 7 profile)
:\>auto_rip -n H:\Users\Corey -u H:\Users\Corey\AppData\Local\Microsoft\Windows -c user_file

xtract all information from all registry hives without using -c switch.
:\>auto_rip -s H:\Windows\System32\config -n H:\Users\Corey -u H:\Users\Corey\AppData\Local\Microsoft\Windows

xtract all information from the SAM, Security, Software, and System hives then store output reports in a specified directory.
:\>auto_rip -s H:\Windows\System32\config -r C:\reports

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```

```
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

     Slot     Start        End          Length       Description
00:  Meta     0000000000   0000000000   0000000001   Primary Table (#0)
01:  -----    0000000000   0000002047   0000002048   Unallocated
02:  00:00    0000002048   0081922047   0081920000   Unknown Type (0x27)
03:  00:01    0081922048   0082536447   0000614400   Unknown Type (0x27)
04:  00:02    0082536448   0976771119   0894234672   NTFS (0x07)
05:  -----    0976771120   0976773167   0000002048   Unallocated
```

```
90423 1377153865|FILE|WKS26A03||.A.B [2583] c:/Windows/System32/migwiz/replacementmanifests/RasmanService-ReplV2.man
90424 1371567743|FILE|WKS26A03||M... [2583] c:/Windows/System32/migwiz/replacementmanifests/RasmanService-ReplV2.man
90425 1391559705|FILE|WKS26A03||MACB [96] c:/Windows/System32/migwiz/replacementmanifests/Rasppp-Repl.man ($FILE_NAME)
90426 1391559709|FILE|WKS26A03||..C. [2838] c:/Windows/System32/migwiz/replacementmanifests/Rasppp-Repl.man
90427 1377153865|FILE|WKS26A03||.A.B [2838] c:/Windows/System32/migwiz/replacementmanifests/Rasppp-Repl.man
90428 1371567743|FILE|WKS26A03||M... [2838] c:/Windows/System32/migwiz/replacementmanifests/Rasppp-Repl.man
90429 1391559705|FILE|WKS26A03||MACB [96] c:/Windows/System32/migwiz/replacementmanifests/RasRqs-Repl.man ($FILE_NAME)
90430 1391559709|FILE|WKS26A03||..C. [1814] c:/Windows/System32/migwiz/replacementmanifests/RasRqs-Repl.man
90431 1377153865|FILE|WKS26A03||.A.B [1814] c:/Windows/System32/migwiz/replacementmanifests/RasRqs-Repl.man
90432 1371567744|FILE|WKS26A03||M... [1814] c:/Windows/System32/migwiz/replacementmanifests/RasRqs-Repl.man
90433 1391559705|FILE|WKS26A03||MACB [102] c:/Windows/System32/migwiz/replacementmanifests/RasServer-Repl.man ($FILE_NAME)
90434 1391559709|FILE|WKS26A03||..C. [27444] c:/Windows/System32/migwiz/replacementmanifests/RasServer-Repl.man
90435 1377153865|FILE|WKS26A03||.A.B [27444] c:/Windows/System32/migwiz/replacementmanifests/RasServer-Repl.man
90436 1371567744|FILE|WKS26A03||M... [27444] c:/Windows/System32/migwiz/replacementmanifests/RasServer-Repl.man
```

6. Static Malware Analysis of njRAT:

Objective:
The objective of this analysis is to examine the njRAT malware sample using static techniques to understand its structure, functionality, and potential malicious behavior without executing the file.

Attribute	Value	Notes
File Name	b02c3cc8be9da55c745c60521c8dc3d2c0a1494cfc8127b6785b58496a779383.exe	njRAT sample
MD5 / SHA-256	SHA-256: B02C3CC8BE9DA55C745C60521C8DC3D2C0A1494CFC8127B6785B58496A779383	Verified
File Type / Packer	PE32 (.NET MSIL), Not Packed	Normal entropy
Suspicious Strings	malicious.com, keylogging APIs, registry paths	Indicators of C2
VirusTotal Detections	63 / 71 engines	Highly malicious
YARA Rule Name	njrat_detection_rule	

YARA Rule Snippet

```
rule njrat_detection_rule
{
meta:
description = "Detects njRAT sample"
author = "Analyst"
strings:
   $s1 = "This program cannot be run in DOS mode"
   $s2 = "GetAsyncKeyState"
   $s3 = "malicious.com"

condition:
   2 of ($s*)
}
```

In the static analysis of the sample, it was found that njRAT is a malware program written in .NET and has malicious properties. This file is not packed and has several suspicious strings, some of which include URLs and APIs used for keystrokes logging and system monitoring. Hash check performed in VirusTotal showed 63/71 detections, and it clearly shows that the file is indeed malicious. From the imported functions, one can see that njRAT is capable of logging keystrokes and other user activity.

7. Dynamic Malware Analysis Practice:

Behaviour Type	Detail	MITRE ATT&CK
Process Created	Suspicious child process	T1059
Registry Modified	Startup persistence key added	T1547
Network Connection	Outbound connection to C2 IP	T1071
File Dropped	Malicious file in system path	T1105

Analysis of the njRAT malware sample via dynamic analysis revealed several malicious activities. The sample created child processes and set up persistence by altering the registry. Furthermore, the sample was creating child processes, dropped files in system folders, and was trying to communicate over the network. Such network communications were identified through traffic analysis of the sample and included suspicious DNS queries and external communications. This behavior is consistent with techniques used as per the MITRE ATT&CK framework such as input capture, persistence, and application layer communication.

8. Capstone Project: End-to-End Digital Forensic Investigation

Evidence

The following evidence was collected and verified using SHA-256 hashing:

- USB Drive (16GB)
- Disk Image (raw(dd) format)
- Network Capture File (.pcap)
- Suspicious Executable File
- Windows Registry Hives

Chain of Custody procedures were followed to ensure integrity and admissibility.

Analysis

Disk image analysis using Autopsy revealed deleted files containing sensitive keywords such as “confidential” and “password.”

Network traffic analysis using Wireshark identified suspicious outbound connections to unknown IP addresses, indicating possible command-and-control (C2) communication.

Registry analysis using RegRipper showed persistence mechanisms through startup entries and evidence of recently accessed sensitive files.

Static malware analysis revealed suspicious strings, malicious imports, and high detection rates on VirusTotal. Dynamic analysis confirmed malicious behavior, including file modifications, registry changes, and external communication.

Conclusion

The system was compromised, with clear evidence of unauthorized access, persistence, and potential data exfiltration. The presence of malware and suspicious network activity confirms malicious intent.